ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** simplified representation	Simplification.
2	**(b)** well-tested explanation	Explains.
3	**(d)** 2 km	Divide by 1000.
4	**(d)** metre	m.
5	**(a)** a pattern	Regularity.
6	**(c)** 0.5 kg	Divide by 1000.
7	**(b)** kilogram	kg.
8	**(d)** approximate value	Approximate.
9	**(d)** 2 minutes	60 s per minute.
10	**(b)** second	s.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Simplified representation of a real system (2).	2
12	Approximate calculation (2).	2
13	Study of matter and its changes (2).	2
14	Metre (1); Kilogram (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Length: metre (1); Mass: kilogram (1); Time: second (1).	1+1+1
16	Burning (1); Rusting (1); Cooking food (1).	1+1+1
17	Reliability (1); Reduce errors (1); Check the results (1).	1+1+1
	OR: 54 x 5/18 (1); = 15 (1); m/s (1).	1+1+1
18	24 x 60 (1); = 1440 (1); minutes (1).	1+1+1
	OR: Parallax (1); Faulty instrument (1); Careless reading (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Question (1); Hypothesis (1); Experiment (1); Observation (1); Conclusion (1).	1+1+1+1+1
	OR: Physical: no new substance (1); Chemical: new substance (1); Examples of physical (1); Examples of chemical (1); Signs (1).	1+1+1+1+1
20	Atomic model (1); Globe (1); Ball and stick (1); Simplify reality (1); Change with evidence (1).	1+1+1+1+1
	OR: Units give meaning (1); Aircraft fuel error (1); Conversion (1); Standard units (1); Careful use (1).	1+1+1+1+1

Mistakes to avoid:

- Multiplying by $18/5$ instead of $5/18$ (or the reverse) when converting between km/h and m/s; remember that m/s values are always *smaller* numbers.
- Writing a measured value without its unit, or mixing units (cm with m) in one calculation.
- Confusing *law* with *theory*: a law describes a pattern, a theory explains it; a theory does not 'become' a law.
- Taking the mean of the readings as 'the true value'; it is only a more reliable estimate, and a single reading is not 'wrong' just because it differs from others.
- Treating a model as an exact copy of reality, or saying that a model that fails a test should be ignored instead of being improved.
- Mixing up accuracy and precision, and reading the least count of an instrument wrongly (use the smallest division, not the numbered one).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** c	c.
2	**(a)** temperature	Temperature.
3	**(a)** 1000 m	Prefix kilo.
4	**(b)** stopwatch	Time.
5	**(d)** science	Trust.
6	**(d)** 100 cm	Prefix centi.
7	**(b)** balance	Mass.
8	**(a)** tested	Testable.
9	**(d)** 1000 g	kilo.
10	**(a)** metre scale	Length.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3 x 1000 (1); = 3000 m (1).	1+1
12	Burning of wood (1); Rusting of iron (1).	1+1
13	Well-tested explanation supported by evidence (2).	2
14	2 x 1000 (1); = 2000 g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	20 x 18/5 (1); = 72 (1); km/h (1).	1+1+1
16	Medicines (1); Fertilisers (1); Fuels (1).	1+1+1
17	Time to reach school (1); Amount of food (1); Distance to a place (1).	1+1+1
	OR: Law describes (1); Theory explains (1); Both are tested (1).	1+1+1
18	Question (1); Experiment (1); Conclusion (1).	1+1+1
	OR: Observation is what we see (1); Inference is an explanation (1); Inference may be wrong (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Metre m (1); Kilogram kg (1); Second s (1); Kelvin K (1); Standard (1).	1+1+1+1+1
	OR: Make a prediction (1); Design experiment (1); Measure (1); Compare (1); Revise the model (1).	1+1+1+1+1
20	(a) 5000 m (1); (b) 2.5 m (1); (c) 3500 g (1); (d) 7200 s (1); (e) 20 m/s (1).	1+1+1+1+1
	OR: Describes patterns (1); Calculates speed (1); Predicts motion (1); Converts units (1); Language of science (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** repeatable	Repetition.
2	**(c)** 3600 s	60 x 60.
3	**(d)** chemical change	New substance.
4	**(d)** describing patterns	Language of science.
5	**(a)** tested	Testable.
6	**(a)** physical change	State change.
7	**(b)** a simplified picture	Model.
8	**(b)** matter	Matter and changes.
9	**(c)** chemical change	New substance.
10	**(a)** motion	Mechanics.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A testable statement about what will happen (2).	2
12	Statement describing a regular pattern in nature (2).	2
13	3 x 3600 (1); = 10800 s (1).	1+1
14	To compare measurements (1); To avoid confusion (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	To simplify (1); To focus on important factors (1); To make calculations possible (1).	1+1+1
16	450 cm (1); 4500 mm (1); Using 1 m = 100 cm = 1000 mm (1).	1+1+1
17	Curiosity (1); Honesty (1); Patience (1).	1+1+1
	OR: Melting of ice (1); Boiling of water (1); Dissolving sugar (1).	1+1+1
18	km (1); cm (1); mm (1).	1+1+1
	OR: Metre scale (1); Balance (1); Stopwatch (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hypothesis (1); Law (1); Theory (1); Examples (1); Testing (1).	1+1+1+1+1
	OR: Indian physicist (1); Ionisation equation (1); Stellar spectra (1); Used physics laws (1); Applied to stars (1).	1+1+1+1+1
20	0.5 litre per breath (1); 20000 breaths (1); 10000 litres (1); Approximate (1); Useful (1).	1+1+1+1+1
	OR: Complex system (1); Small changes grow (1); Limited data (1); Model errors (1); Better for short term (1).	1+1+1+1+1

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